

[GETNET] Aplicação de Oracle DBRUs ("PSUs") 19.x em sistemas que utilizam datapump import

Introdução

Segue abaixo procedimento recomendado de aplicação de Oracle DBRUs 19.x em sistemas que utilizam o utilitário Oracle Datapump (seção deste email, abaixo: " **Instalação de futuros DBRUs ("PSUs") no RDBMS Oracle 19c com DPBP instalado**").

Público-alvo: executores de DBRUs 19.x em sistemas que utilizam o Oracle Datapump

Histórico sumário: em algum momento do ciclo de vida de produtos RDBMS Oracle 19c, foram introduzidos bugs no datapump import, como registrado no caso do servidor **PAYware SQA (gncasspa00540)** - algures entre os DBRUs 19.23 e 19.24

Referências:

1. **SR 3-42090922161** : Datapump invalid on non-multitenant non-RAC database version 19.28.
2. Data Pump Import (IMPDP) Fails with ORA-39126 ORA-1403 (Doc ID 729346.1).
3. How To Reload Datapump Utility EXPDP/IMPDP (Note 430221.1).
4. Relinking Oracle Home FAQ (Frequently Asked Questions) (Doc ID 1467060.1).
5. Datapatch fails with 'ORA-20001','ORA-04063: package body 'SYS.KUPU\$UTILITIES_INT' has errors' (Doc ID 2934003.1).
6. Data Pump Recommended Proactive Patches For 19.10 and Above ([Doc ID 2819284.1](#)).

Contexto:

Após a aplicação bem-sucedida do DBRU 19.28 em 06/08/2025 no OH RDBMS do IC gncasspa00540, mas mal-sucedida no dicionário ("datapatch") do DB DBLACSTG (PAYware STG), objetos relacionados ao utilitário datapump ficaram inválidos, impossibilitando o uso do datapump, e neste caso específico, para o processo de backup/ILM requerido pelo PAYware.

O DPBP (DataPump Bundle Patch: 35787077, para o DBRU 19.21, conforme referência "f" da seção "Referências" deste email) encontrado lá, instalado pela primeira vez sob demanda de TI-Engenharia em janeiro/2024, foi desinstalado, mas mesmo assim a aplicação do respectivo datapatch do DBRU 19.28 falhava, deixando o DB DBLACSTG com objetos inválidos.

Isto posto, tendo como base a evidência de que o outro banco de dados residente no mesmo IC, o DBLAC (PAYware SQA: DBLACQA), não apresentava erros, tampouco objetos inválidos, nem aplicação de datapatch do DPBP (comportamento esperado, porque este DB DBLACQA é periodicamente refrescoado em sua totalidade a partir do backup full do DB PAYware PRD, cujo dicionário tem registrado o DBRU 19.20), funcionava **sem erros** sob o RDBMS DBRU 19.28 instalado em 06/08/2025, E a necessidade de se prosseguir com o backup/ILM da tabela PRODACQR.OUTGOING_IPM_EXT_OLD, partiu-se para a instalação e configuração de um ***novo*** banco de dados (DB) de stage (STG) para o PAYware, o ORASTG, instalado e configurado sob o RDBMS 19.28.

Entretanto, depois de se instalar e configurar este novo DB ORASTG 19.28, permanecemos com erro ao executar o datapump import, inviabilizando o processo de backup/ILM, o qual é um requisito não-funcional do sistema PAYware da GETNET. Em decorrência disto, tivemos que executar um "relink all" dos binários do Oracle RDBMS (uma vez que foi tentado instalar o DBRU 19.23 sem desinstalar previamente o DPBP for 19.21), e instalar o DPBP para DBRU 19.28, o 38170982 (ref. "f" deste email), sob recomendação do suporte oracle (ref. "a" deste email).

MAS, ao tentar executar o datapatch do DPBP para o DBRU 19.28, o mesmo retornou erro por conta de pacote inválido no DB ORASTG: ou seja, o DPBP que se propõe a resolver os bugs associados aos pacotes inválidos do datapump, NÃO consegue atualizar o dicionário do DB TARGET por conta dos mesmos pacotes inválidos. Para sanar tal situação, foi necessário executar o script \$ORACLE_HOME/rdbms/admin/dpload.sql (cuja versão foi atualizada com a aplicação no binário RDBMS do DPBP para DBRU 19.28).

Depois, ao executar com êxito o dpload.sql, conseguimos executar o datapatch do DPBP para o DBRU 19.28, e efeito esperado, todas as packages associadas ao datapump foram recompiladas e validadas com sucesso. Os detalhes de execução deste procedimento se encontram abaixo, na seção "Procedimento de instalação do DPBP para DBRU 19.28 com packages datapump inválidas".

Como consequência, pudemos avançar com êxito no processo de backup/ILM do PAYware com o novo DB de STG, o ORASTG.

Adicionalmente, o mesmo procedimento foi executado no DB DBLACSTG, igualmente com êxito. Tanto o DBRU 19.28 quanto o DPBP foram registrados com sucesso no dicionário.

Instalação de futuros DBRUs ("PSUs") no RDBMS Oracle 19c com DPBP instalado:

(ref. "a" deste email) *** ATENÇÃO *** Este procedimento **é mandatório** em RDBMS/DB(s) que tenham DPBP instalado previamente. Não se deve instalar novo DBRU sem desinstalar o DPBP legado, tampouco não se deve deixar de instalar o novo DPBP após a instalação bem-sucedida do novo DBRU.

1. Verificar se existe DPBP aplicado no Oracle RDBMS do host (as o.s. Oracle RDBMS user):

```
$ cd $ORACLE_HOME/OPatch
$ ./opatch lsinventory | grep DATAPUMP
Patch description: "DATAPUMP BUNDLE PATCH 19.28.0.0.0"
```

2. Se sim, verificar se o datapatch do DPBP foi aplicado em cada um dos DBs hospedados no host (sqlplus):

```
col host_name format a20
set lines 230
col comp_name format a40
set pages 900
col description format a60
col action_time format a30
col comments format a30
col action format a9
col namespace format a9
col id format 99999999
col database_status format a20
col version format a15
col status format a10
```

```
SQL> select patch_id,action,status,action_time,description from dba_registry_sqlpatch;
```

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
----------	--------	--------	-------------	-------------

```
37847857 APPLY SUCCESS 04/09/2025 19:43 OJVM RELEASE UPDATE: 19.28.0.0.250715 (37847857)
37960098 APPLY SUCCESS 04/09/2025 19:54 Database Release Update : 19.28.0.0.250715 (37960098)
38170982 APPLY SUCCESS 10/09/2025 20:51 DATAPUMP BUNDLE PATCH 19.28.0.0.0
```

3. Se sim, desinstalar o DPBP do(s) DB(s) respectivos e do RDBMS do host. Ver documentação de desinstalação do DPBP respectivo.
4. Instalar nova versão do DBRU (p.ex. "19.30").
5. Instalar nova versão do DPBP correspondente à nova versão do DBRU no RDBMS do host. Ver ref. "f" deste email e documentação de instalação do DPBP respectivo.
6. Instalar datapatch do DPBP novo nos DBs que previamente tinham DPBP instalado.

Procedimento de instalação do DPBP para DBRU 19.28 com packages datapump inválidas:

Erros ao se executar o dpimp (datapump import):

A) Logo após a criação e configuração do novo DB STG, o ORASTG, com o DBRU 19.28:

```
Processing object type TABLE_EXPORT/TABLE/TABLE
ORA-39126: Worker unexpected fatal error in KUPW$WORKER.PUT_DDLS [TABLE:"PRODACQR"."OUTGOING_IFM_EXT_OLD"
]
ORA-01403: no data found
ORA-06512: at "SYS.DBMS_SYS_ERROR", line 95
ORA-06512: at "SYS.KUPW$WORKER", line 12623
ORA-06512: at "SYS.DBMS_SYS_ERROR", line 95
ORA-06512: at "SYS.KUPW$WORKER", line 26274
ORA-06512: at "SYS.KUPW$WORKER", line 25771
----- PL/SQL Call Stack -----
   object      line      object
   handle      number     name
7000103fded01d0 33543 package body SYS.KUPW$WORKER.WRITE_ERROR_INFORMATION
7000103fded01d0 12651 package body SYS.KUPW$WORKER.DETERMINE_FATAL_ERROR
7000103fded01d0 26338 package body SYS.KUPW$WORKER.PUT_DDLS
7000103fded01d0 5281 package body SYS.KUPW$WORKER.LOAD_METADATA
7000103fded01d0 13791 package body SYS.KUPW$WORKER.DISPATCH_WORK_ITEMS
7000103fded01d0 2439 package body SYS.KUPW$WORKER.MAIN
7000103bdf4b698 2 anonymous block
DBMS_METADATA.SET_TRANSFORM_PARAM
DBMS_METADATA.SET_TRANSFORM_PARAM
KUPW:
KUPW: In procedure BUILD_OBJECT_STRINGS - non-base info
DBMS_METADATA.CONVERT
DBMS_METADATA.CONVERT
KUPW: l_cur = 1
KUPW: l_pos = 0
KUPW:
KUPW: In procedure DETERMINE_FATAL_ERROR with ORA-01403: no data found
ORA-06512: at "SYS.KUPW$WORKER", line 26274
ORA-06512: at "SYS.KUPW$WORKER", line 25771
Job "SYS"."JOB_IMP_EXCHANGE_20250901" stopped due to fatal error at Sat Sep 6 13:03:42 2025 elapsed 0 00:
00:04
```

b) erro após aplicação do doc. id de referência ao item "b" deste email, e a instalação do DPBP for RUDB 19.28, mas com erro na aplicação de datapatch:

```
UDI-04063: operation generated ORACLE error 4063
ORA-04063: package body "SYS.DBMS_DATAPUMP" has errors
ORA-06508: PL/SQL: could not find program unit being called: "SYS.DBMS_DATAPUMP"
ORA-06512: at line 1
```

c) erro ao consultar a view opatch_xml_inv:

ORASTG DB: (database which is subject of this SR)

```
08:30:38 SQL> select * from opatch_xml_inv;
ERROR:
ORA-29913: error in executing ODCIEXTTABLEOPEN callout
ORA-04063: ORA-04063: package body "SYS.KUPU$UTILITIES" has errors
ORA-06508: PL/SQL: could not find program unit being called:
"SYS.KUPU$UTILITIES"
ORA-06512: at line 1
has errors
```

no rows selected

Elapsed: 00:00:00.09

08:30:39

--

DBLACSTG DB:

```
08:31:14 SQL> select * from opatch_xml_inv;
ERROR:
ORA-29913: error in executing ODCIEXTTABLEOPEN callout
ORA-04063: ORA-04063: package body "SYS.KUPU$UTILITIES" has errors
ORA-06508: PL/SQL: could not find program unit being called: "SYS.KUPU$UTILITIES"
ORA-06512: at line 1
has errors
```

no rows selected

Elapsed: 00:00:00.09

08:31:15

d) erro alert.log:

ORA-04063: ORA-04063: package body "SYS.KUPU\$UTILITIES" has errors

Objetos que ficaram inválidos após aplicação do doc. id de referência ao item "b" deste email, e a instalação do DPBP for RUDB 19.28, mas com erro na aplicação de datapatch: (e permaneceram inválidos mesmo após execução do script utlrp.sql)

REFDATE	OWNER	OBJECT_NAME	SUBOBJECT_NAME	OBJECT_TYPE	CREATED	TIMESTAMP	LAST_DDL_STATUS
06-SEP-25	SYS	DBMS_DATAFOPH		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:15	06-SEP-25 INVALID
06-SEP-25	SYS	DBMS_METADATA		PACKAGE BODY	19-MAY-19	2025-09-06:14:59:15	06-SEP-25 INVALID
06-SEP-25	SYS	DBMS_METADATA_UTIL		PACKAGE BODY	19-MAY-19	2025-09-06:14:59:36	06-SEP-25 INVALID
06-SEP-25	SYS	DBMS_FLOUTS		PACKAGE BODY	19-MAY-19	2025-09-06:14:59:14	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_10_1_TABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:59:39	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_10_2_TABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:59:13	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_IOTABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:59:13	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_MTABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:59:00	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_MTPART_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:52	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_MTPART_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:42	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_IOTAPART_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:51	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_VTABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:05	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_VTABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:58	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_TABLE_DATA_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:32	06-SEP-25 INVALID
06-SEP-25	SYS	KUS_USER_VIEW		VIEW	19-MAY-19	2025-09-06:14:58:00	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUEUE		PACKAGE	19-MAY-19	2025-09-06:14:58:54	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUEUE		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:43	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUEUE_INT		PACKAGE	19-MAY-19	2025-09-06:14:58:18	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUEUE_INT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:55	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUE_INT		PACKAGE	19-MAY-19	2025-09-06:14:58:35	06-SEP-25 INVALID
06-SEP-25	SYS	KUPC\$QUE_INT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:45	06-SEP-25 INVALID
06-SEP-25	SYS	KUPDSDATA		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:40	06-SEP-25 INVALID
06-SEP-25	SYS	KUPDSDATA_INT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:40	06-SEP-25 INVALID
06-SEP-25	SYS	KUPFSFILE		PACKAGE	19-MAY-19	2025-09-06:14:58:55	06-SEP-25 INVALID
06-SEP-25	SYS	KUPFSFILE		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:43	06-SEP-25 INVALID
06-SEP-25	SYS	KUPMSMCP		PACKAGE	19-MAY-19	2025-09-06:14:58:43	06-SEP-25 INVALID
06-SEP-25	SYS	KUPMSMCP		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:56	06-SEP-25 INVALID
06-SEP-25	SYS	KUPFSPROC		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:46	06-SEP-25 INVALID
06-SEP-25	SYS	KUPUSUTILITIES		PACKAGE BODY	19-MAY-19	2025-09-06:15:03:57	06-SEP-25 INVALID
06-SEP-25	SYS	KUPUSUTILITIES_INT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:42	06-SEP-25 INVALID
06-SEP-25	SYS	KUPVFT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:38	06-SEP-25 INVALID
06-SEP-25	SYS	KUPVFT_INT		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:50	06-SEP-25 INVALID
06-SEP-25	SYS	KUPVWORKER		PACKAGE BODY	19-MAY-19	2025-09-06:14:58:46	06-SEP-25 INVALID

34 rows selected.

Posição ANTES da instalação do DPBP for DBRU 19.28:

```
DPBP for DBRU 19.28: 38170982
```

```
$ ./opatch lsinventory | grep 37960098
Patch 37960098 : applied on Wed Aug 06 11:56:32 BRT 2025
Patch description: "Database Release Update : 19.28.0.0.250715 (37960098)"
```

```
$ ./opatch lsinventory | grep OPatch
OPatch version : 12.2.0.1.47
```

```
$ date
```

```
Wed Sep 10 19:48:15 -03 2025
```

```
$ pwd
```

```
/u01/app/oracle/product/19.17.0/dbhome_1/rdbms/admin
```

```
$ ls -ltr dpload.sql
```

```
-rw-r--r-- 1 oracle oinstall 74459 Oct 06 2023 dpload.sql
```

```
$ date
```

```
Wed Sep 10 19:48:24 -03 2025
```

```
./opatch lsinventory | grep 38170982
```

```
<none>
```

Descompactação do DPBP e verificação de compatibilidade:

```
$ unzip -d <PATCH_TOP_DIR> p38170982_1928000DBRU_Generic.zip
```

```
$ cd /u02/oracle/patches/38170982
```

```
$ cd <PATCH_TOP_DIR>/38170982
```

```
$ opatch prereq CheckConflictAgainstOHWithDetail -ph ./
```

```
$ opatch prereq CheckConflictAgainstOHWithDetail -ph ./
```

```
Oracle Interim Patch Installer version 12.2.0.1.47
```

```
Copyright (c) 2025, Oracle Corporation. All rights reserved.
```

```
PREREQ session
```

```
Oracle Home : /u01/app/oracle/product/19.17.0/dbhome_1
```

```
Central Inventory : /u01/app/oraInventory
```

```
from : /u01/app/oracle/product/19.17.0/dbhome_1/oraInst.loc
```

```
OPatch version : 12.2.0.1.47
```

```
OUI version : 12.2.0.7.0
```

```
Log file location : /u01/app/oracle/product/19.17.0/dbhome_1/cfgtoollogs/patch/patch2025-09-10_20-25-03PM_1.log
```

```
Invoking prereq "checkconflictagainstohwithdetail"
```

```
Prereq "checkConflictAgainstOHWithDetail" passed.
```

```
OPatch succeeded.
```

Instalação do DPBP for DBRU 19.28 no Oracle RDBMS do host:

<shutdown nos DBs associados ao RDBMS do host: DBLACQA (SID DBLAC), DBLACSTG e ORASTG>

```
$ cd <PATCH_TOP_DIR>/38170982
```

```
$ opatch apply
```

```
$ opatch apply
```

```
Oracle Interim Patch Installer version 12.2.0.1.47
```

```
Copyright (c) 2025, Oracle Corporation. All rights reserved.
```

```
Oracle Home : /u01/app/oracle/product/19.17.0/dbhome_1
```

```
Central Inventory : /u01/app/oraInventory
```

```
from : /u01/app/oracle/product/19.17.0/dbhome_1/oraInst.loc
```

```
OPatch version : 12.2.0.1.47
```

```
OUI version : 12.2.0.7.0
```

```
Log file location : /u01/app/oracle/product/19.17.0/dbhome_1/cfgtoollogs/patch/patch2025-09-10_20-25-58PM_1.log
```

```
Verifying environment and performing prerequisite checks...
```

```
OPatch continues with these patches: 38170982
```

```
Do you want to proceed? [y|n]
```

```
y
```

```
User Responded with: Y
```

```
All checks passed.
```

```
Backing up files...
```

```
Applying interim patch '38170982' to OH '/u01/app/oracle/product/19.17.0/dbhome_1'
```

```
Patching component oracle.rdbms.dbscripts, 19.0.0.0.0...
```

```
Patching component oracle.rdbms, 19.0.0.0.0...
```

```
Patch 38170982 successfully applied.
```

```
Log file location: /u01/app/oracle/product/19.17.0/dbhome_1/cfgtoollogs/patch/patch2025-09-10_20-25-58PM_1.log
```

```
OPatch succeeded.
```


Posição pós-instalação DPBP no RDBMS do host:

```
$ opatch lsinventory | grep 38170982
Patch 38170982 : applied on Wed Sep 10 20:27:44 BRT 2025

$ date
Wed Sep 10 20:30:11 -03 2025
$ pwd
/u01/app/oracle/product/19.17.0/dbhome_1/rdbms/admin
$ ls -ltr dpload.sql
-rw-r--r-- 1 oracle oinstall 74670 Jul 12 02:25 dpload.sql
$ date
Wed Sep 10 20:30:24 -03 2025
```

Restart dos DBs:

```
SQL> startup (DBLAC)
srvctl start database -d DBLACSTG
srvctl start database -d ORASTG
```

Aplicação de datapatch nos DBs que tinham DPBP - DBLACSTG e ORASTG (mal-sucedida):

```
ORASTG:

$ ./datapatch
SQL Patching tool version 19.28.0.0.0 Production on Wed Sep 10 20:38:29 2025
Copyright (c) 2012, 2025, Oracle. All rights reserved.

Log file for this invocation: /u01/app/oracle_19c/cfgtoollogs/sqlpatch/sqlpatch_57410914_2025_09_10_20_38_29/sqlpatch_invocation.log

Connecting to database...OK
Gathering database info...done
Bootstrapping registry and package to current versions...done
Error: prereq checks failed!
verify_queryable_inventory returned ORA-20001: Latest xml inventory is not loaded into table
Prereq check failed, exiting without installing any patches.

Please refer to MOS Note 1609718.1 and/or the invocation log
/u01/app/oracle_19c/cfgtoollogs/sqlpatch/sqlpatch_57410914_2025_09_10_20_38_29/sqlpatch_invocation.log
for information on how to resolve the above errors.

SQL Patching tool complete on Wed Sep 10 20:38:32 2025

20:42:11 SQL> select * from OPATCH_XML_INV ;
ERROR:
ORA-29913: error in executing ODCIEXTTABLEOPEN callout
ORA-04063: ORA-04063: package body "SYS.KUPUSUTILITIES" has errors
ORA-06508: PL/SQL: could not find program unit being called: "SYS.KUPUSUTILITIES"
ORA-06512: at line 1
has errors

Executar o script dpload.sql:

ref. How To Reload Datapump Utility EXPDP/IMPDP (Note 430221.1)

@$ORACLE_HOME/rdbms/admin/dpload.sql
```

Reexecutar o datapatch do DPBP para DBRU 19.28 nos DBs que tenham previamente o DPBP (DBLACSTG e ORASTG):

```
--run datapatch again:

$ cd $ORACLE_HOME/OPatch
$ ./datapatch
SQL Patching tool version 19.28.0.0.0 Production on Wed Sep 10 20:51:15 2025
Copyright (c) 2012, 2025, Oracle. All rights reserved.

Log file for this invocation: /u01/app/oracle_19c/cfgtoollogs/sqlpatch/sqlpatch_61670008_2025_09_10_20_51_15/sqlpatch_invocation.log

Connecting to database...OK
Gathering database info...done
Bootstrapping registry and package to current versions...done
Determining current state...done

Current state of interim SQL patches:
Interim patch 34411846 (OJVM RELEASE UPDATE: 19.17.0.0.221018 (34411846)):
  Binary registry: Not installed
  SQL registry: Not installed
Interim patch 35354406 (OJVM RELEASE UPDATE: 19.20.0.0.230718 (35354406)):
  Binary registry: Not installed
  SQL registry: Not installed
Interim patch 35648110 (OJVM RELEASE UPDATE: 19.21.0.0.231017 (35648110)):
  Binary registry: Not installed
  SQL registry: Not installed
Interim patch 35787077 (DATAFUMP BUNDLE PATCH 19.21.0.0.0):
  Binary registry: Not installed
  SQL registry: Not installed
Interim patch 36199232 (OJVM RELEASE UPDATE: 19.23.0.0.240416 (36199232)):
  Binary registry: Not installed
  SQL registry: Not installed
Interim patch 37847857 (OJVM RELEASE UPDATE: 19.28.0.0.250715 (37847857)):
  Binary registry: Installed
  SQL registry: Applied successfully on 04-SEP-25 07:43:23.600052 PM
Interim patch 38170982 (DATAFUMP BUNDLE PATCH 19.28.0.0.0):
  Binary registry: Installed
  SQL registry: Not installed

Current state of release update SQL patches:
Binary registry:
  19.28.0.0.0 Release_Update 250708204013: Installed
SQL registry:
  Applied 19.28.0.0.0 Release_Update 250708204013 successfully on 04-SEP-25 07:54:11.591610 PM

Adding patches to installation queue and performing prereq checks...done
Installation queue:
  No interim patches need to be rolled back
  No release update patches need to be installed
  The following interim patches will be applied:
    38170982 (DATAFUMP BUNDLE PATCH 19.28.0.0.0)

Installing patches...
Patch installation complete. Total patches installed: 1

Validating logfiles...done
Patch 38170982 apply: SUCCESS
Logfile: /u01/app/oracle_19c/cfgtoollogs/sqlpatch/sqlpatch_61670008_2025_09_10_20_51_15/sqlpatch_invocation.log
SQL Patching tool complete on Wed Sep 10 20:52:51 2025
```

Verificação de validade de objetos de DB e de componentes Oracle:

```
select /*+ parallel(a,5) */ sysdate reldate,owner,object_name,subobject_name,object_type,created,timestamp,last_ddl_time,status
from dba_objects a
where status <> 'VALID'
order by owner,object_name,object_type;

no rows selected

col host_name format a20
set lines 230
col comp_name format a40
set pages 900
col description format a60
col action_time format a30
col comments format a30
col action format a9
col namespace format a9
col id format 99999999
col database_status format a20
col version format a15
col status format a10

select name db_name,db_unique_name,database_role,open_mode,switchover_status from v$database;
select inst_id,instance_name,host_name,status,version,logins,database_status from gv$instance order by 1 ;
select comp_name,version,status from dba_registry order by comp_name;
select patch_id,action,status,action_time,description from dba_registry_sqlpatch;
select * from dba_registry_history order by action_time;

COMP_NAME                                VERSION                                STATUS
-----
JServer JAVA Virtual Machine              19.0.0.0.0                            VALID
OLAP Analytic Workspace                   19.0.0.0.0                            VALID
Oracle Database Catalog Views             19.0.0.0.0                            VALID
Oracle Database Java Packages             19.0.0.0.0                            VALID
Oracle Database Packages and Types        19.0.0.0.0                            VALID
Oracle Database Vault                    19.0.0.0.0                            VALID
Oracle Label Security                    19.0.0.0.0                            VALID
Oracle Multimedia                        19.0.0.0.0                            VALID
Oracle OLAP API                          19.0.0.0.0                            VALID
Oracle Real Application Clusters           19.0.0.0.0                            OPTION OFF
Oracle Text                              19.0.0.0.0                            VALID
Oracle Workspace Manager                  19.0.0.0.0                            VALID
Oracle XDK                                19.0.0.0.0                            VALID
Oracle XML Database                      19.0.0.0.0                            VALID
Spatial                                  19.0.0.0.0                            VALID

15 rows selected.

Elapsed: 00:00:00.01
20:55:10 SQL>
  PATCH_ID ACTION      STATUS      ACTION_TIME      DESCRIPTION
-----
37847857 APPLY        SUCCESS     04-SEP-25 07:43:23.600052 FM OJVM RELEASE UPDATE: 19.28.0.0.250715 (37847857)
37960098 APPLY        SUCCESS     04-SEP-25 07:54:11.531610 FM Database Release Update : 19.28.0.0.250715 (37960098)
38170982 APPLY        SUCCESS     10-SEP-25 08:43:59.488992 FM DATAFUMP BUNDLE PATCH 19.28.0.0.0

3 rows selected.
```